



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

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Experiment 4

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Q1. Consider a relation R having attributes as R(ABCD), functional dependencies are given below:

$AB \rightarrow C$, $C \rightarrow D$, $D \rightarrow A$

Identify the set of candidate keys possible in relation R. List all the set of prime and non prime attributes.

Given: R(A,B,C,D) with FDs

- $AB \rightarrow C$
 - $C \rightarrow D$
 - $D \rightarrow A$
- $(AB)^+ = ABCD$
 $(BC)^+ = ABCD$
 $(BD)^+ = ABCD$

So the Candidate Keys are : { **AB, BC, BD** }

Prime attributes: { A, B, C, D }

Non-prime attributes: none $\rightarrow \emptyset$

Normal form :

BCNF: No

- Condition: For every FD $X \rightarrow Y$, X must be a superkey.
- $C \rightarrow D$: C is not a superkey $\rightarrow$ violates.
- $D \rightarrow A$: D is not a superkey $\rightarrow$ violates.

3NF: Yes

- Condition: For every FD $X \rightarrow Y$, either X is a superkey **or** every attribute in Y is prime.
- $AB \rightarrow C$: AB is a key $\rightarrow$ OK.

- $C \rightarrow D$: C not a superkey, but D is prime $\rightarrow$ OK.
- $D \rightarrow A$ not a superkey, but A is prime $\rightarrow$ OK.

Normal Form : **3NF**

Q2. Relation R(ABCDE) having functional dependencies as :

$A \rightarrow D$, $B \rightarrow A$, $BC \rightarrow D$, $AC \rightarrow BE$

Identify the set of candidate keys possible in relation R. List all the set of prime and non prime attributes.

Given: R(A,B,C,D,E) with FDs

- $A \rightarrow D$
- $B \rightarrow A$
- $BC \rightarrow D$
- $AC \rightarrow BE$

Closures:

- $(BC)^+ = \{A, B, C, D, E\}$
- $(AC)^+ = \{A, B, C, D, E\}$

So the Candidate Keys are: { BC, AC }

Prime attributes: { A, B, C }

Non-prime attributes: { D, E }

Normal form:

BCNF: No

- Condition: For every FD $X \rightarrow Y$, X must be a superkey.
- $A \rightarrow D$: A not a superkey $\rightarrow$ violates.
- $B \rightarrow A$: B not a superkey $\rightarrow$ violates.
-

3NF: No

- Condition: For every FD $X \rightarrow Y$, either X is a superkey or every attribute in Y is prime.
- $A \rightarrow D$: A not a superkey, D is non-prime $\rightarrow$ violates $\rightarrow$ not in 3NF.
-

2NF: No

Normal Form: 1NF

Q3. Consider a relation R having attributes as R(ABCDE), functional dependencies are given below:

$B \rightarrow A, A \rightarrow C, BC \rightarrow D, AC \rightarrow BE$

Identify the set of candidate keys possible in relation R. List all the set of prime and non prime attributes.

Given: R(A,B,C,D,E) with FDs

- $B \rightarrow A$
- $A \rightarrow C$
- $BC \rightarrow D$
- $AC \rightarrow BE$

Closures:

- ☐ $(A)^+ = \{A, C, B, E, D\} = \{A, B, C, D, E\}$
- ☐ $(B)^+ = \{B, A, C, E, D\} = \{A, B, C, D, E\}$
- ☐ $(C)^+ = \{C\}$
- ☐ $(D)^+ = \{D\}$
- ☐ $(E)^+ = \{E\}$

Candidate Keys: { A, B }

Prime attributes: { A, B }

Non-prime attributes: { C, D, E }

BCNF: Yes

- $B \rightarrow A$: B is a superkey $\rightarrow$ Ok
- $A \rightarrow C$: A is a superkey $\rightarrow$ Ok

So on.. all LHS are superkeys

Normal Form: **BCNF**

Q4. Consider a relation R having attributes as R(ABCDEF), functional dependencies are given below:

$A \rightarrow BCD, BC \rightarrow DE, B \rightarrow D, D \rightarrow A$

Identify the set of candidate keys possible in relation R. List all the set of prime and non prime attributes.

Given: $R(A,B,C,D,E,F)$ with FDs

- $A \rightarrow B C D$
- $BC \rightarrow D E$
- $B \rightarrow D$
- $D \rightarrow A$

Closures:

- $(A)^+ = \{A, B, C, D, E\}$
- $(B)^+ = \{A, B, C, D, E\}$
- $(D)^+ = \{A, B, C, D, E\}$
- $(F)^+ = \{F\}$
- $(AF)^+ = \{A, B, C, D, E, F\} \rightarrow \text{key}$
- $(BF)^+ = \{A, B, C, D, E, F\} \rightarrow \text{key}$
- $(DF)^+ = \{A, B, C, D, E, F\} \rightarrow \text{key}$

Candidate Keys: $\{ AF, BF, DF \}$

Prime attributes: $\{ A, B, D, F \}$

Non-prime attributes: $\{ C, E \}$

BCNF: No

- $A \rightarrow BCD$: A is not a superkey $\rightarrow$ violates.

3NF: No

- $A \rightarrow C$ (part of $A \rightarrow BCD$): A not a superkey and C is non-prime $\rightarrow$ violates.

2NF: No

- AF is a key, but $A \rightarrow C$ (C non-prime) is a partial dependency on part of key AF $\rightarrow$ violates.

1NF: Yes — attributes assumed atomic.

Normal Form: 1NF

Q5. Designing a student database involves certain dependencies which are listed below:

$X \rightarrow Y$

$WZ \rightarrow X$

$WZ \rightarrow Y$

$Y \rightarrow W$

$Y \rightarrow X$

$Y \rightarrow Z$

Given: attributes $\{W, X, Y, Z\}$ with FDs

- $X \rightarrow Y$
- $WZ \rightarrow X$
- $WZ \rightarrow Y$
- $Y \rightarrow W$
- $Y \rightarrow X$
- $Y \rightarrow Z$

Closures:

- $(Y)^+ = \{Y, W, X, Z\} = \text{all attributes} \rightarrow Y \text{ is a key.}$
- $(X)^+ = \{X, Y, W, Z\} = \text{all attributes} \rightarrow X \text{ is a key.}$
- $(WZ)^+ = \{W, Z, X, Y\} = \text{all attributes} \rightarrow WZ \text{ is a key.}$

Candidate Keys: $\{X, Y, WZ\}$

Prime attributes: $\{W, X, Y, Z\}$

Non-prime attributes: $\emptyset$

BCNF: Yes

- Every FD has LHS that is a superkey $(X, Y, WZ) \rightarrow$ satisfies BCNF.

Normal Form: **BCNF**

Q6. Debix Pvt Ltd needs to maintain database having dependent attributes ABCDEF. These attributes are functionally dependent on each other for which functionally dependency set F given as:

$\{A \rightarrow BC, D \rightarrow E, BC \rightarrow D, A \rightarrow D\}$ Consider a universal relation $R1(A, B, C, D, E, F)$ with functional dependency set F , also all attributes are simple and take atomic values only. Find the highest normal form along with the candidate keys with prime and non-prime attribute.

Given: $R(A, B, C, D, E, F)$ with FDs

- $A \rightarrow BC$
- $D \rightarrow E$
- $BC \rightarrow D$
- $A \rightarrow D$

Closures:

- $(A)^+ = \{A, B, C, D, E\}$
- $(AF)^+ = \{A, B, C, D, E, F\} \rightarrow \text{key}$
- $(BC)^+ = \{B, C, D, E\}$
- $(D)^+ = \{D, E\}$
- $(F)^+ = \{F\}$

Candidate Keys: $\{AF\}$

Prime attributes: $\{A, F\}$

Non-prime attributes: $\{B, C, D, E\}$

BCNF: No

- $A \rightarrow BC / A \rightarrow D$: A not a superkey $\rightarrow$ violates.

3NF: No

- $A \rightarrow BC$: A not a superkey and B, C are non-prime $\rightarrow$ violates.

2NF: No

- AF is a key, $A \rightarrow BC$ (BC non-prime) is a partial dependency on part of the key $\rightarrow$ violates.

1NF: Yes — attributes atomic.

Normal Form: 1NF



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